5.6 Antiderivatives; Differential Equations

Name _____ Period ____ Date _____

AP Calculus 5.6 · Student sheet

I can…

- find all the antiderivatives of a function.
- solve a differential equation with an initial condition.
- find velocity and position from acceleration.

Vocabulary · fill in the missing word

Antiderivative — a _____ F with $F'(x) = f(x)$

General antiderivative — $F(x) + C$, all the _____ of f

Differential equation — an equation that contains a _____, like $\frac{dy}{dx} = f(x)$

Initial condition — a _____ value like $y(0) = 4$, used to find C

Part 1 · Reverse it

Find a function whose derivative is given.

f(x)	A function F with F′ = f	Another one
$2x$		
$\cos x$		
e^x		
$3x^2$		

Part 2 · Notes and examples (fill in each note, then work the example)

All antiderivatives: if $F' = f$, then every antiderivative of f is $F(x)$ + _____.

Example 1

Find all the antiderivatives of $f(x) = \cos x$.

Power Rule for antiderivatives: $x^n \rightarrow$ _____ + C, for $n \neq -1$.

Example 2

Find all the antiderivatives of $f(x) = 6x^2 - 4x + 5$.

Example 3

Find all the antiderivatives of $f(x) = e^x + \frac{1}{x}$, $x > 0$.

Initial condition: substitute the known value to find _____.

Example 4

Solve $\frac{dy}{dx} = 2x + 3$ with $y(0) = 4$.

Motion: v is an antiderivative of a, and s is an antiderivative of _____.

Example 5

$a(t) = 6t$, $v(0) = 2$, $s(0) = 1$. Find $s(2)$.

Part 3 · Partner practice

Partner 1

Find all the antiderivatives of $f(x) = 4x^3 - 2x$.

Partner 2

(AP® Practice) Find the solution of $f'(x) = \frac{3}{x}$, $x > 0$, with $f(e) = 7$.

(A) $f(x) = 3 \ln x + 4$ (B) $f(x) = 3 \ln x - 4$

(C) $f(x) = \ln x + 6$ (D) $f(x) = 3 \ln x + 7$